Prasannakumar M

Data Scientist

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Data-driven professional with 6 years of experience leveraging Python, SQL, Pandas and NumPy for data preparation, exploratory analysis, automation and problem solving. Hands on experience in feature engineering, machine learning, model evaluation and optimization, data visualization through end-to-end Data Science projects. Certified in Advanced Data Science, with practical exposure to ML, DL and model deployment.

Skills

- **Programming & Databases:** Python, SQL, MongoDB
- **Data Science:** Pandas, NumPy, EDA, Data Cleaning, Data Transformation, Feature Engineering
- **Machine Learning:** Scikit-learn, Regression, Classification, Clustering, Model Evaluation
- **Deep Learning:** TensorFlow, PyTorch, CNN, Transfer Learning
- **Data Visualization:** PowerBI, Matplotlib, Seaborn, Plotly
- **Deployment & Tools:** Streamlit, FastAPI, Jupyter Notebook, Google Colab, Git

Experience

Senior Automation Engineer (Jul 2024 – Nov 2025) – *Indium Software, Chennai*

- Managed test execution across Android, Kindle, iOS and Web platforms, generating operational execution data used to monitor quality, performance and recurring failures.
- Analyzed execution data and operational KPIs to identify failure patterns, investigate recurring issues and support process improvements through data-driven insights.
- Developed Python-based automation for execution-report parsing, data consolidation and selective re-execution, reducing redundant regression cycles and improving operational efficiency.

Senior Catalog Associate (Nov 2020 – Jul 2024) – *Amazon, Chennai*

- Analyzed large-scale product quality metrics and operational data to identify data inconsistencies and recurring quality patterns across multiple marketplaces.
- Developed Python and Excel-based automation using Pandas, OpenPyXL and VBA for data validation, processing and reporting, reducing report preparation time by nearly 90%.
- Supported catalog operations for 350K+ products while maintaining 98%+ catalog accuracy through data validation, quality analysis and process improvements.

Technical Support Associate (Jan 2020 – Jun 2020) – *Altruist Technologies Pvt Ltd, Bengaluru*

Customer Support Associate (Jun 2019 – Nov 2019) – *Amazon, Coimbatore*

Education

Bachelor of Engineering, *KSR Institute for Engineering and Technology* 2015 - 2019

Advanced Professional Certification in Data Science, *GUVI Geek Network Pvt. Ltd.* 2026

Projects

Telecom Customer Churn Prediction | *Python, Pandas, NumPy, Scikit-learn, Feature Engineering, ML*

- Performed data preprocessing and feature engineering on a **5.8K+** record telecom dataset, creating features to capture customer usage and service patterns such as AvgMonthlySpend, TenureBucket and ServiceCount.
- Built and evaluated **Logistic Regression, Random Forest, Gradient Boosting** and **XGBoost** models using **accuracy, precision, recall, F1-score** and **ROC-AUC**.
- Applied preprocessing pipelines with **OneHotEncoder, StandardScaler** & stratified train-test splitting, then deployed the trained churn model through a **FastAPI** prediction API supporting customer retention analysis.

Content Monetization Modeler | *Python, Pandas, Scikit-learn, Lasso Regression, Streamlit*

- Developed a machine learning model to estimate content monetization potential using **122K+** records.
- Performed data preprocessing, exploratory analysis and feature engineering, **comparing regression models** to identify the best-performing approach.
- Selected Lasso Regression with an R^2 of 0.953 and **deployed the model through Streamlit** to support content monetization estimation.

Fish Image Classification | *Python, TensorFlow, CNN, Transfer Learning, Deep Learning*

- Developed a deep learning pipeline to classify & identify 11 fish species using transfer learning on **6K+** images.
- Trained and compared MobileNetV2, ResNet50, VGG16, EfficientNetB0 and InceptionV3 models.
- Achieved 99.45% validation accuracy with ResNet50 and saved the trained model for image-based fish classification and inference.